OSINACHI CHUKWUJAMA

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PROFILE

First-year photonics master's student with a First Class degree in Mechanical Engineering, seeking a 3-month electro-optical engineering internship from June 2027. I bring hands-on experience in industrial equipment maintenance and reporting, a grounding in physical optics, and several years of software work that make me at ease with IT tools, test procedures and technical documentation. I am keen to apply this to the setup, reliability testing and calibration of infrared test equipment.

EDUCATION

Erasmus Mundus Joint Master, Intelligent Photonics (iPSRS) September 2026 – Present
M1, Optics, Image, Vision and Multimedia, Université Jean Monnet, Saint-Étienne, France
Relevant coursework: Physical and Fourier optics (diffraction, 4f systems, interferometry), digital image processing.

Bachelor of Engineering, Mechanical Engineering November 2017 – August 2023
Federal University of Technology, Owerri (FUTO), Nigeria. Major: Industrial and Production Engineering
CGPA: 4.56/5.00 (First Class), best graduating student in the option.

SKILLS

Optics & maths	Physical and Fourier optics, probability; translating derivations into working numerical code
Hands-on & industrial	Preventive maintenance of production machines, maintenance reporting and equipment tracking, CAD and 3D printing of custom parts, embedded hardware (Jetson Nano)
IT tools	Python (proficient), MATLAB, JavaScript; Go and PHP (intermediate); Linux, Git, CI/CD and automated testing
Languages	English: C1 (IELTS 8.0)

LABORATORY, RESEARCH AND PROJECTS

Spatial Light Modulator (SLM) Laboratory Practical 2026
IPSRS M1 coursework *Université Jean Monnet*

- Generated diffraction gratings and Fresnel lens phase patterns in MATLAB and displayed them on an SLM connected to the setup.
- Observed and analysed the resulting diffraction patterns, and computed the gratings' spatial frequencies from measurements in the acquisition software.

Surveillance Drone with an Object Detection System August 2023
Undergraduate Final Project, supervised by Prof. O.O. Obiukwu *MEE, FUTO*

- Built an end-to-end system: data collection, YOLOv5 model training (70% mAP) and live inference on a drone's camera stream.
- Follow-up (March 2024): deployed the model on an NVIDIA Jetson Nano with TensorRT, cutting processing time by 44%.

Optics Simulation Playground In progress
Personal project, pre-zeemax.vercel.app

- 2D ray-tracing tool with finite-size lenses, being extended into a Fourier optics simulator.

WORK EXPERIENCE

Rite Foods March 2024 – June 2024
Mechanical Technician Trainee (graduate internship) *Lagos, Nigeria*

- Carried out daily routine maintenance on baking and packaging machines, keeping downtime at or below 20%.
- Wrote daily maintenance reports so that equipment and supplies were properly tracked.
- Designed and 3D printed replacement parts for machines with no immediate spare available.

Generative CAD December 2021 – June 2022
Robotics & 3D Printing Industrial Trainee *Lagos, Nigeria*

- Gathered customer requirements, designed 3D models and printed prototypes (PLA, ABS) to validate them.

· Developed ROS path planning for robots, simulated in Gazebo and RViz.

Dstrct Group March 2025 – May 2026
Backend Engineer / Team Lead *Lagos, Nigeria*

· Technical lead on a large enterprise platform (GraphQL, NestJS, PostgreSQL); turned client requirements into engineering tasks.
· Built an automated pipeline generating formal DOCX/PDF documents from database records.

PUBLICATIONS

Secure and Resilient Industrial IoT Architectures for Smart Manufacturing: A Comprehensive Review, JERR (2025). doi:10.9734/jerr/2025/v27i61548

Clinical Applications of 3D-Printed Medical Implants, JOPIR. doi:10.69613/22dn3m33

Quantum Computing in Artificial Intelligence: A Review of Quantum Machine Learning Algorithms, Path of Science. doi:10.22178/pos.117-25

HONOURS, TEAMWORK AND COMMUNICATION

First Class graduate, FUTO (2023), 96th percentile in the department; PTDF Scholarship (2018).

Microsoft Certified Trainer and former **Microsoft Learn Student Ambassador**: trained over 300 students in software and cloud technologies.

Director of Research, Mechanical Engineering, FUTO: ran hands-on training in 3D printing, microcontrollers and machine learning for 50+ students and staff. Co-founder of Hack4FUTO.